KERALA POLYTECHNIC · REVISION 2026

COURSE 1254 | SEMESTER I | PRACTICUM | 4 CREDITS | 60 CONTACT HOURS

Computer Applications in Management

Complete handbook covering computer hardware, operating systems, data processing, networks — with Malayalam support, interactive calculators, diagrams, and exam preparation.

PROGRAMME Computer Application & Business Management	COURSE TITLE Computer Applications in Management
COURSE CODE 1254	TEACHING SCHEME 3:0:2:0 (L:T:P:R)
CREDITS 4	CIE MARKS 40
ESE MARKS 100	SEMESTER I

OFFICIAL SOURCE

Source Declaration

This handbook is based on the official Kerala Polytechnic Revision 2026 syllabus for Course **1254 — Computer Applications in Management**.

📄 **Source integrity:** The syllabus PDF was used as the authoritative source. No external materials were consulted. The handbook follows the official course structure, learning outcomes, and assessment scheme exactly as specified.

⚠️ **Verified inconsistencies:** The PDF contains some minor formatting inconsistencies in the teaching scheme table (the header "Self-learning (SS)/Week" appears with a colon but no value in the row) and the practical assessment table has some column alignment variations. These have been interpreted according to the most consistent reading across the document. The handbook presents the information as it is most coherently understood from the official source.

REV2026 | SOURCE: SITTTR | COURSE 1254

AT A GLANCE

Course Dashboard

Key statistics and structure of Course 1254 — Computer Applications in Management.

4

Modules

Covering computer basics, OS, data processing, and networks

60

Contact Hours

Total instructional hours across theory and practical

4

Credits

Practicum course with 3:0:2:0 teaching scheme

140

Total Marks

40 CIE + 100 ESE

Teaching Scheme/Week	Self-learning/Week	Total Hours/Semester	Credits	CIE		ESE		Total
3:0:2:0	5.5	60	4	20	20	60	40	140

Teaching and Assessment Scheme — official structure

📋 Pre-requisites

Basic computer knowledge at the Secondary Education level.

🎯 Course Type

Practicum — hands-on laboratory work with theory components.

LEARNING PATH

How to Use This Handbook

Follow this four-step roadmap to get the most out of your study.

1

Understand

Read each module's theory, definitions, and concepts. Use the Malayalam notes for clarity.

2

Visualise

Study the SVG diagrams and network topology illustrations. Watch the animated generators.

3

Apply

Work through the calculators, practical procedures, and self-learning activities.

4

Revise

Use the Q&A, model paper, revision cards, and glossary to cement your knowledge.

WHAT YOU WILL LEARN

Course Outcomes

Official course objectives and student learning outcomes mapped to Bloom's Taxonomy.

🎯 Course Objectives

2. Lay foundation in operating systems, data processing, data communication, networks, assembling, and software installation.
3. Explain data communication concepts and identify network components in digital environments.
4. Identify different types of computer networks and compare topologies for organisational requirements.

📊 CO–PO Mapping

Course Outcomes mapped to Program Outcomes (PO1–PO11). Correlation: 1=Low, 2=Medium, 3=High.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	–	–	2	–	–	–	–	–	–
CO2	3	2	2	–	3	–	–	–	–	–	–
CO3	3	2	–	2	2	–	–	–	–	–	–
CO4	3	2	–	2	2	–	–	–	–	–	–

CO	Course Outcome	Bloom's Level
CO1	Explain the basics of computer system and identify the components	Apply
CO2	Understand the types and functions of the operating systems and apply the knowledge to install various system and application software	Apply
CO3	Define Data processing, identify the communication elements and types of data transmission media	Apply
CO4	Identify and illustrate various computer networks and topologies	Apply

Course Outcomes with Bloom's Taxonomy Level

SYLLABUS MAP

Curriculum Coverage

Every module, topic, and subtopic from the official syllabus is covered in this handbook.

Module	Title	Major Topics	L	P	CO
I	Basics of computer system	Introduction, Characteristics, Generations, Components, Memory, Hardware/Software, Types of Software, Languages, Translators, Algorithm, Flowchart	12	0	CO1
I	Components identification & assembling	Motherboard, memory, HDD, CMOS, SMPS, power connector, keyboard connector, Desktop & Laptop assembling, checking system configuration	0	6	CO1
II	Operating system	Definition, Functions, Layers, Types, Techniques, Popular OS, Mobile OS	11	0	CO2

Module	Title	Major Topics	L	P	CO
II	Installation of various OS and application software	Windows, Linux, Application software installation	0	8	CO2
III	Data processing and Communication	Data Processing, File Management, DBMS, Communication elements, Modes, Transmission media, Modems, Wireless	11	0	CO3
III	Data processing & demonstration of cables	Data processing, file types, guided media, modems, wifi, UTP crimping	0	8	CO3
IV	Computer Networks and Topologies	Networks definition, advantages, types, topologies, components, Internet basics, Cloud Storage	11	0	CO4
IV	Basic network troubleshooting and demonstrating network components	Troubleshooting, printer connection, network connectivity, search engines	0	8	CO4

Module-wise coverage with instructional hours and mapped CO

HOW MODULES CONNECT

Concept Roadmap

Each module builds on the previous one, creating a coherent learning journey from hardware to networks.

📘 Module I

Computer Basics — Hardware, software, memory, languages, and assembling.

📗 Module II

Operating Systems — OS functions, types, installation, and mobile OS.

📙 Module III

Data Processing & Communication — File management, DBMS, transmission media, modems.

📕 Module IV

Computer Networks — Topologies, components, internet basics, cloud storage.

💡 **Learning flow:** Start with Module I to understand the machine itself, then Module II to see how software controls it, Module III to handle data and communication, and Module IV to connect everything in a network.

MODULE I

Basics of Computer System & Components Identification

Understanding the fundamental building blocks of a computer — from hardware components to software, memory, languages, and assembling.

Theory: 12 hours | Practical: 6 hours | CO: CO1 | Bloom: Apply

Introduction to Computers — Characteristics & Generations

What is a Computer? A computer is an electronic device that processes data, following a set of instructions (a program) to produce meaningful information. It accepts input, processes it, stores it, and produces output.

Characteristics of Computers

- **Speed:** Computers can process millions of instructions per second. Measured in MIPS (Million Instructions Per Second) or clock speed (GHz).
- **Accuracy:** Computers produce error-free results as long as the input and instructions are correct. Errors are typically due to human mistakes (garbage in, garbage out).
- **Storage:** Computers can store vast amounts of data permanently (secondary storage) and temporarily (primary memory).
- **Versatility:** Computers can perform a wide variety of tasks — from calculations to graphics, gaming, and data analysis.
- **Diligence:** Computers can work for hours without fatigue, performing repetitive tasks with consistent accuracy.
- **Automation:** Once programmed, computers can operate automatically without human intervention.

Generations of Computers

Generation	Technology	Key Features	Examples
First (1940s–1956)	Vacuum Tubes	Large, expensive, consumed huge power, magnetic drums for memory	ENIAC, UNIVAC
Second (1956–1963)	Transistors	Smaller, more reliable, less power, magnetic core memory	IBM 1401, CDC 1604
Third (1964–1971)	Integrated Circuits (ICs)	Multiple transistors on a single chip, keyboards/monitors introduced	IBM 360, PDP-8
Fourth (1971–Present)	Microprocessors	Thousands of ICs on one chip, personal computers, GUI, networking	IBM PC, Apple Macintosh
Fifth (Present & Beyond)	Artificial Intelligence, Quantum Computing	Parallel processing, AI, natural language, quantum bits	AI systems, Quantum computers

Computer Generations — key characteristics

> **Malayalam support:** കമ്പ്യൂട്ടർ എന്നാൽ ഡാറ്റ പ്രോസസ്സ് ചെയ്യുന്ന ഒരു ഇലക്ട്രോണിക് ഉപകരണമാണ്. വേഗത, കൃത്യത, സംഭരണ ശേഷി, വൈവിധ്യം എന്നിവയാണ് പ്രധാന സവിശേഷതകൾ. തലമുറകൾ വാക്വം ട്യൂബുകൾ മുതൽ ട്രാൻസിസ്റ്ററുകൾ, IC-കൾ, മൈക്രോപ്രൊസസ്സറുകൾ, AI വരെ പരിണമിച്ചു.

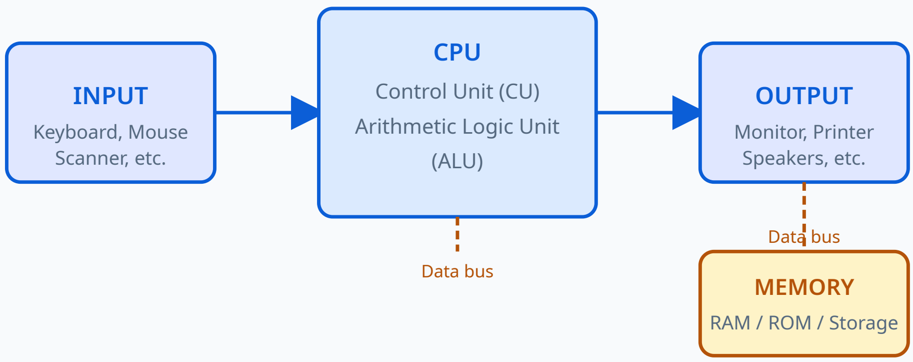
Components of Computer Systems

A computer system consists of **hardware** (physical components) and **software** (programs and data). The hardware components work together in a coordinated manner.

Major Components

- **Input Unit:** Keyboard, mouse, scanner, microphone — devices that feed data into the computer.
- **Central Processing Unit (CPU):** The "brain" of the computer. Contains the Control Unit (CU) and Arithmetic Logic Unit (ALU).
- **Memory Unit:** Stores data and instructions. Includes primary memory (RAM, ROM) and secondary storage (hard drives, SSDs).
- **Output Unit:** Monitor, printer, speakers — devices that display or present processed data.
- **Motherboard:** The main circuit board that connects all components.

Input → CPU (CU + ALU) ↔ Memory → Output



Block diagram showing the flow of data: Input → CPU (CU + ALU) ↔ Memory → Output

> **Malayalam support:** കമ്പ്യൂട്ടർ സിസ്റ്റത്തിന്റെ പ്രധാന ഭാഗങ്ങൾ: ഇൻപുട്ട് യൂണിറ്റ് (കീബോർഡ്, മൗസ്), CPU (തലച്ചോറ്), മെമ്മറി (RAM, ROM, ഹാർഡ് ഡിസ്ക്), ഔട്ട്പുട്ട് യൂണിറ്റ് (മോണിറ്റർ, പ്രിന്റർ). ഇവയെല്ലാം മദർബോർഡ് വഴി ബന്ധിപ്പിച്ചിരിക്കുന്നു.

Computer Memory — Types and Hierarchy

Computer memory is used to store data and instructions. It is organised in a hierarchy from fastest (and most expensive) to slowest (and cheapest).

Types of Memory

- **Primary Memory (Main Memory):**
  - **RAM (Random Access Memory):** Volatile — loses data when power is off. Used for running programs and current data.
  - **ROM (Read Only Memory):** Non-volatile — retains data without power. Contains firmware (BIOS).
  - **Cache Memory:** Very fast, small memory between CPU and RAM. Stores frequently accessed data for speed.
- **Secondary Memory (Storage):**
  - **Hard Disk Drive (HDD):** Magnetic storage, large capacity, slower access.
  - **Solid State Drive (SSD):** Flash-based, faster than HDD, no moving parts.
  - **Optical Media:** CD, DVD, Blu-ray — read/write using lasers.
  - **Removable Media:** USB flash drives, memory cards.

Memory Hierarchy (Speed & Cost)

Memory Type	Speed	Volatility	Cost per GB	Typical Size
Register (CPU internal)	Fastest	Volatile	Highest	Few bytes
Cache (L1/L2/L3)	Very fast	Volatile	Very high	MB
RAM (DRAM/SRAM)	Fast	Volatile	High	GB
SSD (Solid State Drive)	Medium-fast	Non-volatile	Medium	GB–TB
HDD (Hard Disk Drive)	Slow	Non-volatile	Low	TB
Optical (CD/DVD/Blu-ray)	Very slow	Non-volatile	Lowest	GB

Memory hierarchy — from fastest to slowest

⚠️ **Common mistake:** Students often confuse RAM and ROM. Remember: **RAM** is volatile and can be written to; **ROM** is non-volatile and is typically read-only (though some types like EEPROM can be written with special procedures).

> **Malayalam support:** മെമ്മറി രണ്ട് തരത്തിലുണ്ട്: പ്രൈമറി (RAM, ROM, Cache) — വേഗതയേറിയതും ചെലവേറിയതും; സെക്കൻഡറി (HDD, SSD, USB) — സാവധാനവും വിലകുറഞ്ഞതും. RAM വോൾട്ടൈൽ (പവർ പോയാൽ ഡാറ്റ നഷ്ടം), ROM നോൺ-വോൾട്ടൈൽ (പവർ പോയാലും ഡാറ്റ നിലനിൽക്കും).

Hardware, Software & Their Relationship

Hardware refers to the physical components of a computer — the parts you can see and touch. **Software** refers to the programs and data that run on the hardware.

Relation Between Hardware and Software

- Hardware is the **physical body** of the computer; software is the **mind** that tells it what to do.
- Hardware cannot function without software; software cannot run without hardware.
- The operating system acts as the intermediary between the two, managing hardware resources for software applications.

Types of Software

- **System Software:** Manages hardware and provides a platform for applications. Examples: Operating Systems (Windows, Linux), Device Drivers, Utility software.
- **Application Software:** Designed for end-users to perform specific tasks. Examples: MS Office, web browsers, games, Photoshop.
- **Utility Software:** Helps maintain and optimise the system. Examples: Antivirus, Disk Cleanup, Backup software.

Free and Open-Source Software (FOSS)

- **Free Software:** Users can run, copy, distribute, study, change, and improve the software. "Free" as in freedom, not necessarily price.
- **Open-Source Software:** The source code is available for modification and redistribution. Examples: Linux, LibreOffice, Apache, GIMP, VLC.
- **Proprietary Software:** Owned by an organisation; source code is not available. Examples: Microsoft Windows, Adobe Photoshop.

> **Malayalam support:** ഹാർഡ്വെയർ ഭൗതിക ഘടകങ്ങളാണ് (മോണിറ്റർ, കീബോർഡ്, CPU). സോഫ്റ്റ്വെയർ പ്രോഗ്രാമുകളാണ് (OS, MS Office). രണ്ടും പരസ്പരം ആശ്രയിച്ചിരിക്കുന്നു. ഫ്രീ & ഓപ്പൺ സോഴ്സ് സോഫ്റ്റ്വെയർ (FOSS) ഉപയോഗിക്കാൻ, പകർത്താൻ, മാറ്റാൻ സ്വാതന്ത്ര്യം നൽകുന്നു (Linux, LibreOffice).

Computer Languages & Translators

Computers understand instructions in binary (0s and 1s). Different levels of programming languages bridge the gap between human-readable code and machine-executable instructions.

Levels of Programming Languages

- **Machine Language (1GL):** Binary code (0s and 1s). The only language directly understood by the CPU. Fast but difficult for humans to write.
- **Assembly Language (2GL):** Uses mnemonics (e.g., MOV, ADD) instead of binary. Needs an assembler to convert to machine code.
- **High-Level Language (3GL):** Uses English-like syntax (e.g., C, C++, Python, Java). Needs a compiler or interpreter to translate.
- **Fourth-Generation Language (4GL):** Designed for specific applications (e.g., SQL for databases).
- **Fifth-Generation Language (5GL):** Used for artificial intelligence and problem-solving (e.g., Prolog).

Translators

- **Compiler:** Translates the entire program at once, producing an executable file. Fast execution, but errors are found after compilation. Example: GCC (C compiler).
- **Interpreter:** Translates and executes one line at a time. Slower execution, but errors are found immediately. Example: Python interpreter.
- **Assembler:** Translates assembly language into machine code.

Feature	Compiler	Interpreter
Translation	Whole program at once	Line by line
Execution Speed	Faster	Slower
Error Detection	After compilation	During execution
Output	Executable file	No executable file
Example	C, C++	Python, JavaScript

Compiler vs Interpreter

> **Malayalam support:** കമ്പ്യൂട്ടർ ഭാഷകൾ: മെഷീൻ (0/1), അസംബ്ലി (mnemonics), ഹൈ-ലെവൽ (English-like). ട്രാൻസ്ലേറ്ററുകൾ: കമ്പൈലർ (മുഴുവൻ പ്രോഗ്രാമും ഒരുമിച്ച് വിവർത്തനം), ഇന്റർപ്രീറ്റർ (വരി വരിയായി വിവർത്തനം).

Algorithm & Flowchart

Algorithm: A step-by-step procedure to solve a problem. It is written in plain English or pseudo-code. An algorithm must be finite, unambiguous, and effective.

Flowchart: A graphical representation of an algorithm using standard symbols. It visualises the flow of control and data.

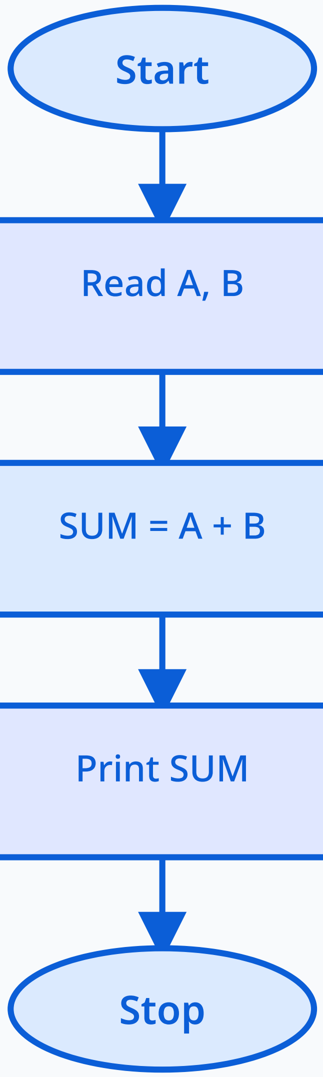
Common Flowchart Symbols

- **Oval (Terminal):** Start / End
- **Parallelogram (Input/Output):** Read / Print
- **Rectangle (Process):** Calculation / Assignment
- **Diamond (Decision):** Condition / Branching
- **Arrows (Flow lines):** Direction of flow

📘 Example: Algorithm to find the sum of two numbers

Step 1: Start
Step 2: Read two numbers A and B
Step 3: Calculate SUM = A + B
Step 4: Print SUM
Step 5: Stop

Start → Read A,B → SUM = A+B → Print SUM → Stop



Flowchart for adding two numbers — shows the sequence of operations.

Malayalam support: അൽഗോരിതം ഒരു പ്രശ്നം പരിഹരിക്കാനുള്ള ഘട്ടം ഘട്ടമായുള്ള നടപടിക്രമമാണ്. ഫ്ലോചാർട്ട് ഇതിന്റെ ഗ്രാഫിക്കൽ പ്രതിനിധാനമാണ്. വ്യത്യസ്ത ആകൃതികൾ വ്യത്യസ്ത പ്രവർത്തനങ്ങളെ സൂചിപ്പിക്കുന്നു: ഓവൽ (ആരംഭം/അവസാനം), പാരലലോഗ്രാം (ഇൻപുട്ട്/ഔട്ട്പുട്ട്), റെക്ടാങ്കിൾ (പ്രോസസ്സിംഗ്), ഡയമണ്ട് (തീരുമാനം).

🔧 Practical: Components Identification & System Assembling

This practical session focuses on identifying computer components and assembling a desktop and laptop system.

Key Components to Identify

- **Motherboard:** The main circuit board. Identify processor sockets, CPU support, power connectors, keyboard connectors, BIOS chip, bus slots, chipset, cache memory, co-processors, and CMOS battery.
- **Memory (RAM):** Identify DIMM/SO-DIMM modules, speed ratings, and capacity.
- **Hard Disk Drive (HDD) / SSD:** Identify connectors (SATA, IDE), power connectors, and storage capacity.
- **SMPS (Switched Mode Power Supply):** Power supply unit that converts AC to DC for the computer.
- **CMOS Battery:** Maintains system configuration and clock when the computer is off.
- **Other connectors:** Power connector, keyboard connector (PS/2 or USB), monitor connector (VGA, HDMI, DVI).

Desktop Assembling Procedure

1. Prepare the case and install the power supply (SMPS).
2. Install the motherboard into the case, aligning with standoffs.
3. Install the CPU into the socket (align the triangle marker).
4. Attach the CPU cooler fan.
5. Install RAM modules into the appropriate slots.
6. Connect power cables from SMPS to motherboard (24-pin), CPU (4/8-pin), and drives.
7. Install HDD/SSD and connect SATA/data cables.
8. Connect front panel connectors (power switch, reset, LEDs, USB, audio).
9. Connect monitor, keyboard, and mouse.
10. Power on and observe the POST screen (Power-On Self-Test) and OS loading.
11. Check system configuration using BIOS/UEFI or system information tools.

Laptop Assembling & Disassembling

- Identify the parts of a laptop: keyboard, trackpad, display, battery, RAM, HDD/SSD, WiFi card, cooling fan, motherboard.
- Practice assembling and disassembling with proper ESD precautions.
- Note the differences from desktop: integrated components, smaller form factor, specialised connectors.

> 🛡️ **Safety note:** Always disconnect the power supply before opening the computer case. Use an anti-static wrist strap to prevent electrostatic discharge (ESD) damage to components. Handle components by their edges, not by the pins or circuits.

Malayalam support: കമ്പ്യൂട്ടർ ഘടകങ്ങൾ തിരിച്ചറിയുന്നതിനും അസംബ്ലിംഗ് ചെയ്യുന്നതിനുമുള്ള പ്രായോഗിക പരിശീലനം. മദർബോർഡ്, CPU, RAM, HDD, SMPS, CMOS ബാറ്ററി എന്നിവ തിരിച്ചറിയുക. ഡെസ്ക്ടോപ്പും ലാപ്ടോപ്പും അസംബിൾ ചെയ്യാൻ പഠിക്കുക. POST സ്ക്രീൻ നിരീക്ഷിക്കുകയും സിസ്റ്റം കോൺഫിഗറേഷൻ പരിശോധിക്കുകയും ചെയ്യുക.

💡 **Module I Summary:** You have learned about computer characteristics, generations, components, memory hierarchy, hardware/software relationship, programming languages, translators, algorithms, flowcharts, and practical component identification and assembling. These form the foundation for all subsequent modules.

MODULE II

Operating Systems — Types, Functions & Installation

Understanding the core software that manages computer hardware and provides a platform for applications — from definitions to installation.

Theory: 11 hours | Practical: 8 hours | CO: CO2 | Bloom: Apply

Definition, Functions & Layers of Operating System

Operating System (OS): A system software that acts as an intermediary between the computer hardware and the user. It manages hardware resources, provides a user interface, and enables application software to run.

Functions of an Operating System

- **Process Management:** Manages the execution of processes (programs). Includes scheduling, creation, and termination of processes.
- **Memory Management:** Allocates and manages memory for programs. Ensures efficient use of RAM.
- **File System Management:** Organises files and directories on storage devices. Manages read/write operations, permissions, and file organisation.
- **Device Management:** Controls I/O devices through device drivers. Handles communication between devices and the CPU.
- **Security & Protection:** Protects user data from unauthorised access. Manages user authentication and permissions.
- **User Interface:** Provides a way for users to interact with the computer (CLI or GUI).
- **Resource Allocation:** Distributes system resources (CPU time, memory, I/O) among processes.

Layers of an Operating System

Layer	Description
Hardware	Physical components: CPU, memory, I/O devices
Kernel	Core of the OS, manages hardware, processes, memory, and files. Runs in privileged mode.
System Services	Utilities, device drivers, file system management, networking stack.
Shell / Command Interpreter	Interface between the user and the OS (CLI or GUI).
User Applications	Application software running on top of the OS (MS Office, browsers, games).

OS Layers — from hardware to user

Malayalam support: ഓപ്പറേറ്റിംഗ് സിസ്റ്റം (OS) ഹാർഡ്വെയറിനും ഉപയോക്താവിനും ഇടയിലുള്ള ഒരു പാലമാണ്. പ്രധാന ധർമ്മങ്ങൾ: പ്രോസസ്സ് മാനേജ്മെന്റ്, മെമ്മറി മാനേജ്മെന്റ്, ഫയൽ സിസ്റ്റം മാനേജ്മെന്റ്, ഡിവൈസ് മാനേജ്മെന്റ്, സുരക്ഷ, യൂസർ ഇന്റർഫേസ്. OS-ന്റെ പാളികൾ: ഹാർഡ്വെയർ → കേർണൽ → സിസ്റ്റം സർവീസസ് → ഷെൽ → ആപ്ലിക്കേഷനുകൾ.

Types of Operating Systems

Operating systems can be classified based on the number of users they support, the processing technique they use, and their response characteristics.

Based on Users

- **Single-User OS:** Designed for one user at a time. Examples: MS-DOS, Windows 95/98.
- **Multi-User OS:** Allows multiple users to use the system simultaneously. Examples: Linux, UNIX, Windows Server.

Based on Processing Technique

- **Batch Processing OS:** Jobs are collected in a batch and processed sequentially. No user interaction during processing. Example: Early mainframes.
- **Time-Sharing OS (Multitasking):** Multiple users share CPU time in small time slices. Gives the illusion of simultaneous execution. Examples: UNIX, Linux, Windows.
- **Real-Time OS (RTOS):** Responds to events in a guaranteed time frame. Used in embedded systems, robotics, and industrial control. Examples: VxWorks, QNX.

Operating System Techniques

- **Multiprogramming:** Multiple programs are loaded into memory, and the CPU switches between them to keep the CPU busy.
- **Multitasking:** Similar to multiprogramming, but includes user interaction. Multiple tasks (processes) appear to run simultaneously.
- **Multiprocessing:** Uses multiple CPUs to execute multiple processes simultaneously. Requires special hardware.

Type	User Interaction	Response Time	Example
Batch Processing	None (offline)	High (hours/days)	Early IBM mainframes
Time-Sharing	Interactive	Low (milliseconds)	UNIX, Linux, Windows
Real-Time	Event-driven	Guaranteed (microseconds)	VxWorks, QNX

Comparison of OS Types

> **Malayalam support:** OS-കളെ ഉപയോക്താക്കളുടെ എണ്ണം (single-user, multi-user), പ്രോസസ്സിംഗ് രീതി (batch, time-sharing, real-time), ടെക്നിക്കുകൾ (multiprogramming, multitasking, multiprocessing) എന്നിവ അടിസ്ഥാനമാക്കി തരം തിരിച്ചിരിക്കുന്നു.

Popular Operating Systems — Linux, Windows & Mobile OS

Microsoft Windows

- **History:** Started with Windows 1.0 (1985). Windows 95 introduced the Start menu. Windows 10/11 are the latest versions.
- **Features:** GUI-based, wide application support, user-friendly, extensive hardware compatibility, regular updates.
- **Editions:** Home, Pro, Enterprise, Education, Server.
- **File System:** NTFS (New Technology File System) — supports large files, security, and compression.

Linux

- **History:** Created by Linus Torvalds in 1991 as a free and open-source alternative to UNIX.
- **Features:** Open-source, free, highly customisable, stable, secure, widely used in servers and embedded systems.
- **Distributions (Distros):** Ubuntu (user-friendly), Fedora, Debian, CentOS, Red Hat Enterprise Linux (RHEL).
- **File System:** ext4, XFS, Btrfs — supports advanced features like journaling and snapshots.

Mobile Operating Systems (MOS)

- **Android:** Based on Linux kernel, open-source, developed by Google. Most widely used mobile OS. Features: customisable, large app ecosystem, supports multiple hardware vendors.
- **iOS:** Developed by Apple, closed-source, runs on iPhone and iPad. Features: highly secure, seamless ecosystem, curated app store, regular updates.
- **Others:** HarmonyOS (Huawei), KaiOS (feature phones), watchOS (Apple Watch).

Key Features of Mobile OS

- Touch-based UI with gesture controls.
- App stores with millions of applications.
- Cloud integration for backup and sync.
- Location services (GPS), sensors (accelerometer, gyroscope).
- Security: biometric authentication (fingerprint, face ID), app permissions.

Future Trends of Mobile OS

- **AI Integration:** On-device AI processing, intelligent assistants.
- **Foldable & Dual-Screen Support:** Optimised for new form factors.
- **Privacy & Security:** Increased focus on user data protection.
- **Cross-Device Integration:** Seamless experience across phone, tablet, laptop, and IoT devices.

Malayalam support: ജനപ്രിയ OS-കൾ: Windows (GUI, wide app support), Linux (open-source, stable, server), Android (open-source, mobile), iOS (closed-source, secure). മൊബൈൽ OS-കളുടെ ഭാവി: AI, ഫോൾഡബിൾ സ്ക്രീനുകൾ, പ്രൈവസി, ക്രോസ്-ഡിവൈസ് ഇന്റഗ്രേഷൻ.

🔧 Practical: Installation of Windows, Linux & Application Software

This practical session covers the installation of Windows and Linux operating systems, as well as various application software packages.

Installing Windows Operating System

1. Insert the Windows installation media (USB drive or DVD).
2. Boot from the installation media (enter BIOS/UEFI to change boot order).
3. Select language, time, and keyboard preferences.
4. Click "Install Now" and enter the product key (if required).
5. Select the operating system edition (e.g., Windows 11 Home/Pro).
6. Accept the license terms.
7. Choose the installation type: "Custom: Install Windows only (advanced)".
8. Select the drive/partition where Windows will be installed.
9. Follow the on-screen prompts for user account creation, privacy settings, and updates.
10. After installation, install device drivers and update the OS.

Installing Linux Operating System (Ubuntu example)

1. Download the Ubuntu ISO file from the official website.
2. Create a bootable USB drive using tools like Rufus or BalenaEtcher.
3. Boot from the USB drive.
4. Select "Try Ubuntu" or "Install Ubuntu".
5. Choose language and keyboard layout.
6. Select "Normal installation" and "Download updates while installing" (optional).
7. Choose "Erase disk and install Ubuntu" or "Something else" for manual partitioning.
8. Select your timezone and create a user account.
9. Wait for the installation to complete and restart the system.
10. Remove the installation media and boot into the new Ubuntu system.

Installing Application Software

- **MS Office:** Run the installer, accept the license, choose the installation type (typical/custom), and complete the setup.
- **Printer Software:** Connect the printer, run the driver installer, or use Windows/Linux built-in drivers.
- **Other Software:** Adobe Reader, web browsers, antivirus, media players, development tools.

> ⚠️ **Common issues:** Ensure you have a valid product key for Windows. For Linux, check hardware compatibility (especially WiFi and graphics drivers). Always back up important data before installing a new OS.

🛡️ **Safety note:** Keep your system plugged in during OS installation to avoid power interruption. Use genuine installation media to avoid malware. After installation, install security updates and antivirus software.

Malayalam support: Windows, Linux OS-കളുടെ ഇൻസ്റ്റലേഷൻ പ്രക്രിയ. ബൂട്ട് മീഡിയ തയ്യാറാക്കുക, BIOS/UEFI-ൽ boot order മാറ്റുക, ഇൻസ്റ്റലേഷൻ വിസാർഡ് പിന്തുടരുക. MS Office, printer drivers തുടങ്ങിയ ആപ്ലിക്കേഷൻ സോഫ്റ്റ്വെയറുകളും ഇൻസ്റ്റാൾ ചെയ്യുക.

💡 **Module II Summary:** You have learned about the definition, functions, and layers of an operating system, different types of OS (single/multi-user, batch, time-sharing, real-time), OS techniques (multiprogramming, multitasking, multiprocessing), popular OS (Windows, Linux, Android, iOS), and practical installation procedures.

MODULE III

Data Processing & Communication

Understanding how data is processed, stored, and transmitted — from file management to communication media and modems.

Theory: 11 hours | Practical: 8 hours | CO: CO3 | Bloom: Apply

Data Processing — Definition, Types & File Management

Data Processing: The process of collecting, manipulating, and converting raw data into meaningful information. It involves input, processing, and output stages.

Types of Data Processing

- **Batch Processing:** Data is collected over time and processed in batches. Used for payroll, billing, and reporting.
- **Real-Time Processing:** Data is processed immediately as it is received. Used for ATM transactions, online booking, and monitoring systems.
- **Online Processing:** Interactive processing where the user waits for a response. Used for data entry and query systems.
- **Distributed Processing:** Data is processed across multiple computers. Used in cloud computing and large-scale systems.

File Management System

- **File:** A collection of related data stored on a storage device.
- **File Types:**
    - **Master File:** Contains permanent data (e.g., employee records, customer details).
    - **Transaction File:** Contains temporary data for processing (e.g., daily sales, daily attendance).
    - **Reference File:** Contains lookup data (e.g., price list, code tables).
    - **Report File:** Contains formatted output for printing or viewing.
- **File Operations:** Create, read, write, update, delete, rename, copy, move, backup.

Database Management System (DBMS)

- **Definition:** A software system that manages the storage, retrieval, and manipulation of structured data.
- **Features:** Data security, integrity, concurrency control, backup and recovery, query language (SQL).
- **Applications:** Banking systems, hospital management, railway reservations, e-commerce, social media.
- **Difference from File System:** DBMS reduces redundancy, ensures data consistency, provides concurrent access, and enforces security rules.

Feature	File System	DBMS
Data Redundancy	High (duplicate data possible)	Low (normalised)
Data Consistency	Difficult to maintain	Automatically maintained
Concurrent Access	Limited or manual locking	Built-in concurrency control
Security	Basic (file permissions)	Granular (user/role-based)
Query Language	No standard	SQL (Structured Query Language)

File System vs DBMS

> **Malayalam support:** ഡാറ്റ പ്രോസസ്സിംഗ്: raw data-യെ meaningful information-ആക്കി മാറ്റുന്ന പ്രക്രിയ. തരങ്ങൾ: batch, real-time, online, distributed. ഫയൽ തരങ്ങൾ: master, transaction, reference, report. DBMS ഘടനാപരമായ ഡാറ്റ മാനേജ് ചെയ്യുന്നു (bank, hospital, railway).

🔧 Practical: Performing Data Processing (Excel Example)

This practical session involves collecting raw data, processing it in Microsoft Excel, and generating formatted reports.

Example: Daily Sales of a Shop for 5 Days

1. **Input:** Enter the daily sales data for 5 days (e.g., product name, quantity sold, price per unit).
2. **Processing:**
   - Calculate total sales for each day (quantity × price).
   - Calculate the average sales over the 5 days.
   - Sort the data by product name or by total sales.
3. **Output:** Generate a formatted report with tables, charts, and summary statistics.

> 📘 **Sample Calculation**
> **Day 1:** Product A — 10 units × ₹50 = ₹500
> **Day 2:** Product A — 12 units × ₹50 = ₹600
> **Day 3:** Product A — 8 units × ₹50 = ₹400
> **Total:** ₹500 + ₹600 + ₹400 = ₹1,500
> **Average:** ₹1,500 ÷ 3 = ₹500 per day

File Operations

- **Create:** New file (e.g., sales_data.xlsx).
- **Rename:** Change file name (e.g., sales_report_2026.xlsx).
- **Copy:** Make a duplicate for backup.
- **Move:** Transfer to a different folder.
- **Delete:** Remove unwanted files.
- **Backup:** Copy to pen drive or cloud storage (Google Drive, OneDrive).

> **Malayalam support:** Excel ഉപയോഗിച്ച് ഡാറ്റ പ്രോസസ്സിംഗ്: റോ ഡാറ്റ നൽകുക, സൂത്രവാക്യങ്ങൾ (total, average, sort) പ്രയോഗിക്കുക, ഫോർമാറ്റ് ചെയ്ത റിപ്പോർട്ട് ജനറേറ്റ് ചെയ്യുക. ഫയൽ ഓപ്പറേഷനുകൾ: create, rename, copy, move, delete, backup.

Data Communication — Elements, Modes & Signals

Data Communication: The exchange of data between two devices via a transmission medium. It is fundamental to networking and the internet.

Elements of a Communication System

- **Sender (Source):** The device that sends the data (e.g., computer, smartphone).
- **Receiver (Destination):** The device that receives the data.
- **Message:** The data being transmitted (text, audio, video, etc.).
- **Transmission Medium:** The physical path through which data travels (wired: copper, fibre; wireless: radio, microwave).
- **Protocol:** A set of rules that governs the communication (e.g., TCP/IP, HTTP).

Modes of Data Communication

- **Simplex:** One-way communication. The sender can only send, and the receiver can only receive. Example: Radio broadcast, TV transmission.
- **Half-Duplex:** Two-way communication, but only one device can send at a time. Example: Walkie-talkie, CB radio.
- **Full-Duplex:** Two-way communication where both devices can send and receive simultaneously. Example: Telephone, video conferencing.

Analog and Digital Signals

- **Analog Signal:** Continuous wave signal that varies in amplitude and frequency. Used in traditional telephone lines, radio, and TV. Susceptible to noise and attenuation.
- **Digital Signal:** Discrete signal with two states (0 and 1). Used in modern computer networks, fibre optics, and digital communication. More immune to noise and easier to process.

Feature	Analog	Digital
Signal Type	Continuous wave	Discrete (0/1)
Noise Susceptibility	High	Low
Processing	Complex	Simple (computers)
Example	Radio, telephone (traditional)	Computer networks, fibre optics

Comparison: Analog vs Digital Signals

Malayalam support: ഡാറ്റ കമ്മ്യൂണിക്കേഷൻ — ഉപകരണങ്ങൾ തമ്മിലുള്ള ഡാറ്റ കൈമാറ്റം. ഘടകങ്ങൾ: sender, receiver, message, medium, protocol. മോഡുകൾ: simplex (one-way), half-duplex (alternate), full-duplex (simultaneous). Signals: analog (continuous) vs digital (discrete 0/1).

Transmission Media — Guided & Unguided

Transmission Media is the physical or wireless path that carries data from sender to receiver. It is broadly classified into guided (wired) and unguided (wireless).

Guided Media (Wired)

- **Twisted Pair Cable:**
  - Two insulated copper wires twisted together to reduce electromagnetic interference.
  - Used in telephone networks, Ethernet (10/100/1000BASE-T).
  - Types: Shielded (STP) and Unshielded (UTP).
  - Advantages: Low cost, flexible, easy to install.
  - Disadvantages: Limited distance, susceptible to interference.
- **Coaxial Cable:**
  - Has a central conductor surrounded by insulation, a braided shield, and an outer jacket.
  - Used in cable TV, broadband internet, and CCTV systems.
  - Advantages: Better shielding, longer distance than twisted pair.
  - Disadvantages: Thicker, more expensive, harder to install.
- **Optical Fibre Cable:**
  - Uses light pulses to transmit data through a glass or plastic core.
  - Used in high-speed internet, backbone networks, and telecommunications.
  - Advantages: Very high bandwidth, immune to electromagnetic interference, long distance (up to 100 km).
  - Disadvantages: High cost, fragile, complex installation.

Unguided Media (Wireless)

- **Radio Waves:** Omni-directional, used in FM/AM radio, Bluetooth, WiFi (2.4 GHz, 5 GHz).
- **Microwaves:** Directional, used in satellite communication, point-to-point links, and radar.
- **Infrared:** Short-range, line-of-sight, used in remote controls and IR communication.
- **Li-Fi (Light Fidelity):** Uses visible light for communication, very high speed.

Media	Bandwidth	Distance	Noise Immunity	Cost
Twisted Pair	Up to 1 Gbps	~100 m	Low	Low
Coaxial	Up to 1 Gbps	~500 m	Medium	Medium
Optical Fibre	100+ Gbps	~100 km	High	High
Wireless (Radio)	~1 Gbps (WiFi 6)	~100 m (indoor)	Low	Medium

Comparison of Transmission Media

Malayalam support: ട്രാൻസ്മിഷൻ മീഡിയ — guided (wired: twisted pair, coaxial, optical fibre) ഉം unguided (wireless: radio, microwave, infrared) ഉം. ഓരോന്നിന്റെയും ഗുണങ്ങളും ദോഷങ്ങളും, bandwidth, distance, cost എന്നിവ താരതമ്യം ചെയ്യുക.

Modems & Wireless Data Connectivity

Modem (Modulator-Demodulator): A device that converts digital signals from a computer into analog signals for transmission over telephone lines (modulation) and converts incoming analog signals back to digital (demodulation).

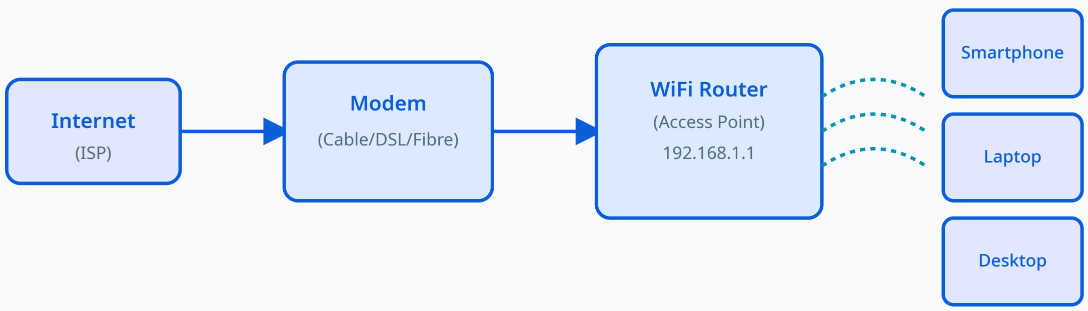
Types of Modems

- **Dial-up Modem:** Uses standard telephone lines (PSTN). Slow speeds (up to 56 kbps). Now mostly obsolete.
- **ADSL Modem:** Uses telephone lines but with higher speeds (up to 24 Mbps). Uses frequency division to separate voice and data.
- **Cable Modem:** Uses coaxial cable TV networks for internet access. Speeds up to 1 Gbps.
- **Fibre Optic Modem (ONT):** Used with fibre-to-the-home (FTTH) connections. Very high speeds (up to 1 Gbps+).
- **Wireless Modem (4G/5G):** Uses cellular networks for internet access. Integrated into smartphones or external dongles.

Basics of Wireless Data Connectivity

- **WiFi (Wireless Fidelity):** Based on IEEE 802.11 standards. Uses radio waves (2.4 GHz and 5 GHz bands).
- **Bluetooth:** Short-range communication (up to 10 m). Used for connecting peripherals (keyboards, headphones).
- **Cellular Networks:** 4G (LTE), 5G — provide mobile internet with wide coverage.
- **Components:** Wireless Access Point (WAP), Router, Network Interface Card (NIC), Antenna.

Internet → Modem → WiFi Router → Smartphone / Laptop / Desktop



A typical WiFi network: Internet → Modem → WiFi Router → Devices (smartphone, laptop, desktop).

Malayalam support: മോഡം — digital to analog (modulation), analog to digital (demodulation) ചെയ്യുന്ന ഉപകരണം. തരങ്ങൾ: dial-up, ADSL, cable, fibre, wireless (4G/5G). WiFi (wireless fidelity) radio waves ഉപയോഗിച്ച് devices-കളെ ബന്ധിപ്പിക്കുന്നു. Router, Access Point, NIC എന്നിവയാണ് പ്രധാന ഘടകങ്ങൾ.

🔧 Practical: Guided Media Demonstration & UTP Cable Crimping

This practical session involves identifying guided media (cables), understanding modems and WiFi components, and performing UTP cable crimping.

Demonstration of Guided Media

- **Twisted Pair Cable (UTP/STP):** Observe the colour-coded wires, the RJ45 connector, and the cable jacket. Understand the T568A and T568B wiring standards.
- **Coaxial Cable:** Observe the central conductor, insulation, braided shield, and outer jacket. Identify the BNC or F-type connector.
- **Optical Fibre Cable:** Observe the glass core, cladding, buffer coating, and connector (SC, LC, ST). Note that it uses light pulses, not electrical signals.
- **Modem & WiFi Components:** Identify different types of modems (ADSL, cable, fibre), wireless access points, and antennas.

UTP Cable Crimping

Tools required: UTP cable, RJ45 connectors, crimping tool, cable stripper, cable tester.

1. Strip about 2.5 cm of the outer jacket from the cable end using the cable stripper.
2. Untwist the four pairs of wires and arrange them according to the T568B standard (or T568A).
3. **T568B wiring order (from left to right, with clip facing down):** Orange/White, Orange, Green/White, Blue, Blue/White, Green, Brown/White, Brown.
4. Cut the wires straight across, leaving about 1.25 cm from the jacket.
5. Insert the wires into the RJ45 connector (ensure the wires go all the way to the end).
6. Place the connector into the crimping tool and crimp firmly.
7. Repeat the process on the other end of the cable.
8. Test the cable using a cable tester to ensure continuity and correct pin configuration.

> ⚠️ **Common mistakes:** Incorrect wire order, wires not reaching the end of the connector, stripping too much jacket, crimping too lightly. Always double-check the T568B/T568A standard before crimping.

> 🛡️ **Safety note:** Use proper tools and handle the cable cutter carefully. Do not pull cables with excessive force. Test cables before using them in a live network.

> **Malayalam support:** UTP കേബിൾ ക്രിംപ്പിംഗ്: T568B വയറിംഗ് സ്റ്റാൻഡേർഡ് പിന്തുടർന്ന് RJ45 കണക്റ്റർ ഘടിപ്പിക്കുക. ക്രിംപ്പിംഗ് ടൂൾ, കേബിൾ സ്ട്രിപ്പർ, കേബിൾ ടെസ്റ്റർ എന്നിവ ഉപയോഗിക്കുക. വയറുകളുടെ ക്രമം: Orange/White, Orange, Green/White, Blue, Blue/White, Green, Brown/White, Brown.

💡 **Module III Summary:** You have learned about data processing types, file management, DBMS, elements of data communication, communication modes (simplex, half-duplex, full-duplex), analog vs digital signals, transmission media (guided and unguided), modems, wireless connectivity, and practical cable crimping.

MODULE IV

Computer Networks & Topologies

Understanding how computers communicate, network topologies, components, internet basics, and cloud storage.

Theory: 11 hours | Practical: 8 hours | CO: CO4 | Bloom: Apply

Computer Communication Networks — Definition & Advantages

Computer Communication Network: A set of interconnected computers and devices that communicate with each other using a common set of protocols. The goal is to share resources, data, and services.

Advantages of Networks

- **Resource Sharing:** Share hardware (printers, scanners), software, and data across the network.
- **Communication:** Email, instant messaging, video conferencing, VoIP.
- **Centralised Data Management:** Data stored on servers can be accessed from any connected device.
- **Cost Efficiency:** Reduces hardware costs by sharing devices and storage.
- **Reliability:** If one device fails, the network can still function (redundancy).
- **Scalability:** Networks can be expanded to include new devices and users.
- **Remote Access:** Users can access network resources from anywhere (VPN, remote desktop).
- **Security:** Centralised security management (firewalls, authentication, encryption).

Malayalam support: കമ്പ്യൂട്ടർ നെറ്റ്വർക്ക് — പരസ്പരം ആശയവിനിമയം നടത്തുന്ന കമ്പ്യൂട്ടറുകളുടെ ഒരു കൂട്ടം. നേട്ടങ്ങൾ: resource sharing (printers, data), communication (email, video), centralised management, cost efficiency, reliability, scalability, remote access, security.

Types of Networks — PAN, LAN, MAN, WAN

Networks are classified based on their size, geographic coverage, and the technology used.

- **PAN (Personal Area Network):**
  - Covers a very small area (a few meters).
  - Used for connecting personal devices: smartphone, laptop, headphones, smartwatch.
  - Technologies: Bluetooth, USB, Infrared.
  - Example: Connecting a Bluetooth headset to a phone.
- **LAN (Local Area Network):**
  - Covers a limited area: a room, a building, a campus.
  - Typically uses Ethernet (wired) and WiFi (wireless).
  - High speed, low latency, private ownership.
  - Example: A school computer lab, office network, home WiFi.
- **MAN (Metropolitan Area Network):**
  - Covers a city or a metropolitan area.
  - Combines multiple LANs in a city.
  - Examples: Cable TV networks, city-wide WiFi, municipal networks.
- **WAN (Wide Area Network):**
  - Covers a large geographic area (country, continent, global).
  - Connects multiple LANs and MANs.
  - Examples: The Internet, corporate networks connecting branch offices.
  - Technologies: Leased lines, satellite, fibre optics, cellular networks.

Type	Size	Speed	Example	Technology
PAN	Few meters	Low–Medium	Bluetooth headset	Bluetooth, USB
LAN	Building/Campus	High (1–10 Gbps)	School computer lab	Ethernet, WiFi
MAN	City	Medium–High	Cable TV network	Fibre, microwave
WAN	Country/Global	Varies	The Internet	Satellite, fibre

Comparison: PAN, LAN, MAN, WAN

> **Malayalam support:** നെറ്റ്വർക്ക് തരങ്ങൾ: PAN (personal — Bluetooth), LAN (local — office/school), MAN (metropolitan — city), WAN (wide — internet). വലിപ്പം, വേഗത, സാങ്കേതികവിദ്യ എന്നിവയിൽ വ്യത്യാസമുണ്ട്.

Network Topologies — Bus, Ring, Mesh, Star

Network Topology: The arrangement or physical layout of devices (nodes) and connections in a network. Each topology has its own advantages and disadvantages.

Bus Topology

- All devices are connected to a single central cable (bus).
- Data travels in both directions along the bus.
- Advantages: Simple, cost-effective, easy to extend.
- Disadvantages: Single point of failure (cable break), limited distance, performance degrades with more devices.

Ring Topology

- Each device is connected to two others, forming a ring.
- Data travels in one direction around the ring.
- Advantages: Predictable performance, no collisions.
- Disadvantages: Single break breaks the entire ring, difficult to troubleshoot.

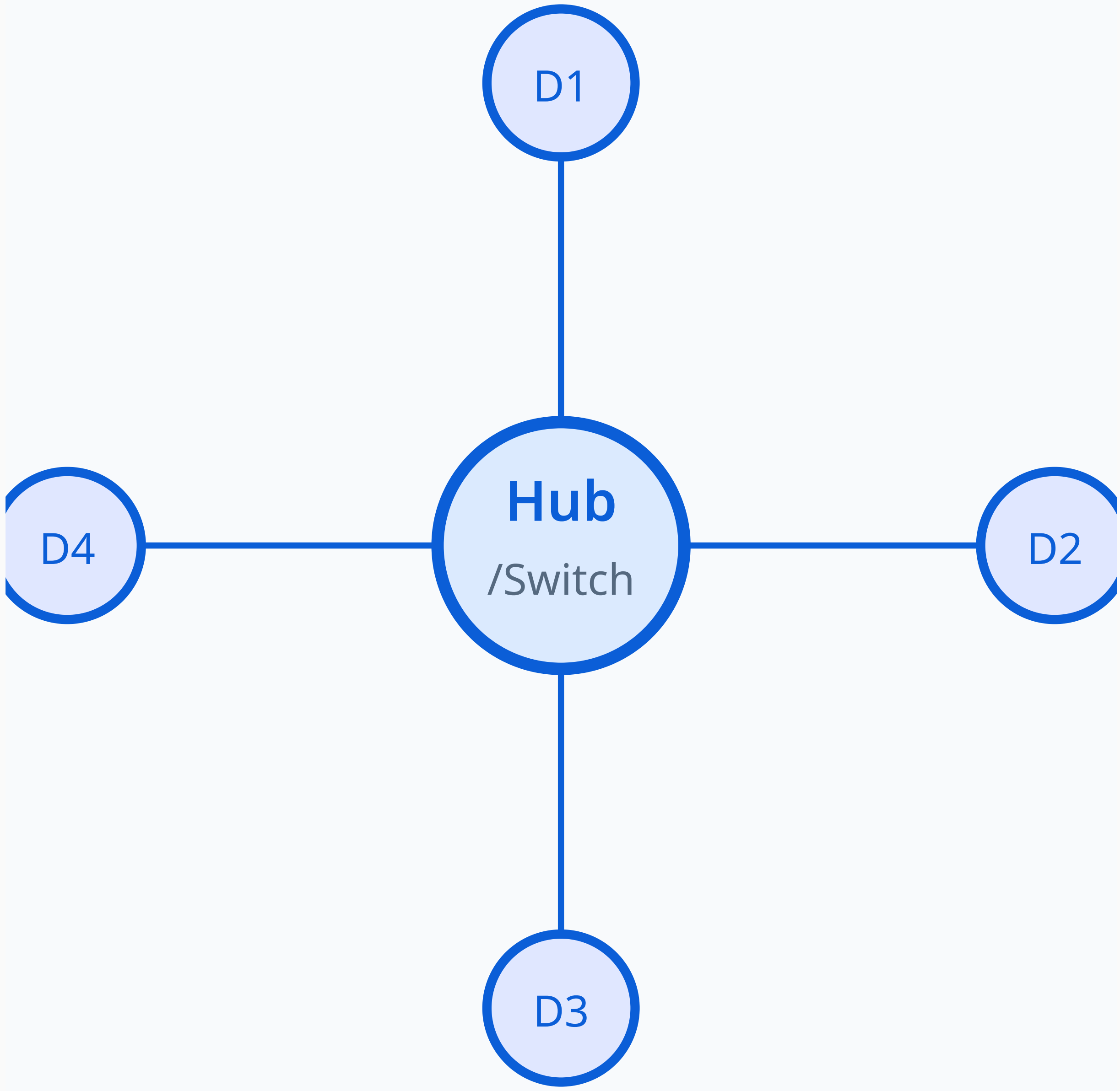
Mesh Topology

- Every device is connected to every other device (full mesh) or partially connected (partial mesh).
- Advantages: High redundancy, multiple paths, robust.
- Disadvantages: Expensive, complex cabling, difficult to manage.

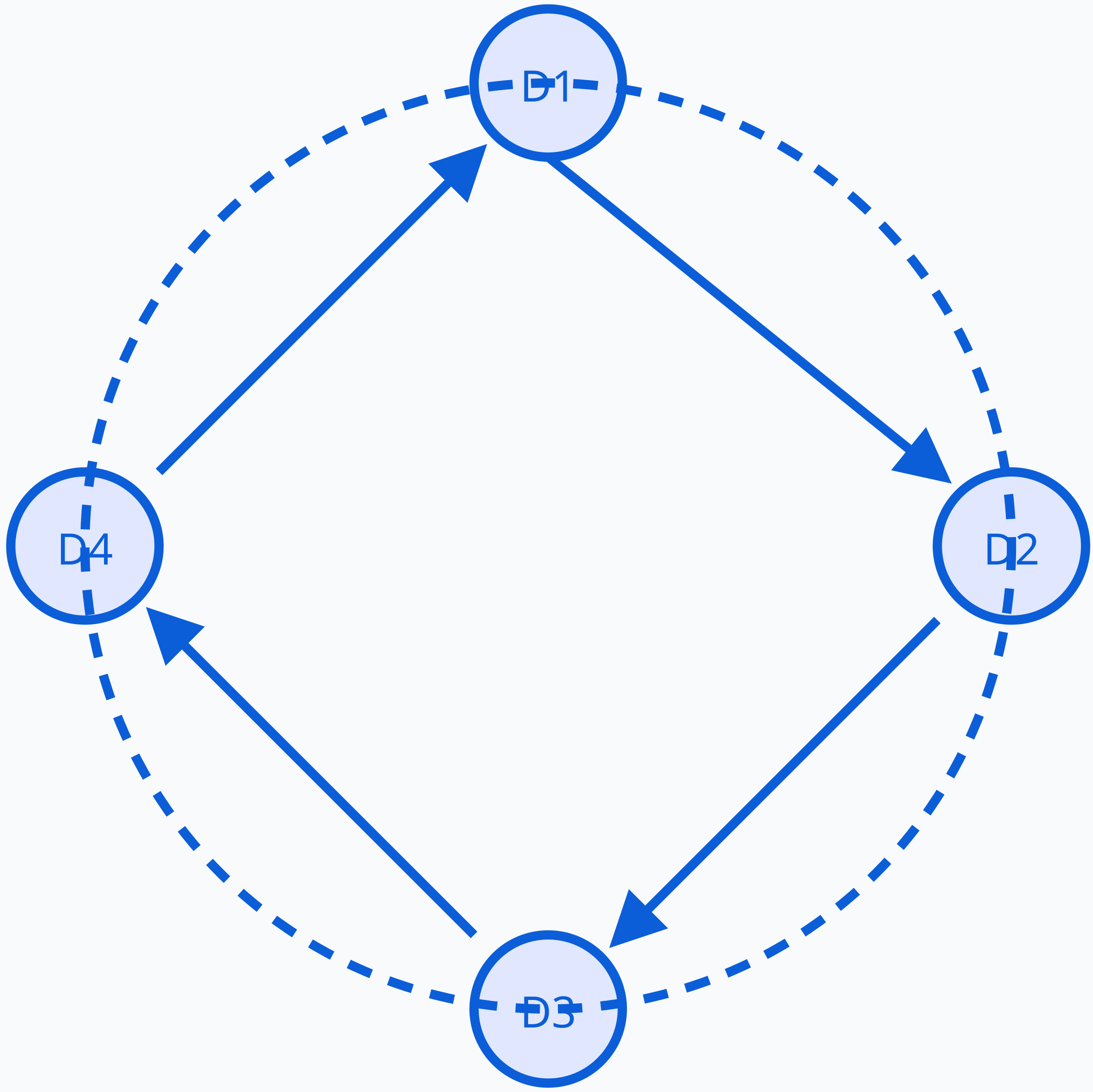
Star Topology

- All devices are connected to a central hub or switch.
- Advantages: Easy to manage, centralised control, failure of one device does not affect others.
- Disadvantages: Single point of failure (central hub/switch), high cabling cost.

Central hub/switch connected to multiple devices



Devices connected in a ring, data flows in one direction



Topology	Advantages	Disadvantages	Applications
Bus	Simple, low cost	Single point of failure, limited scalability	Small networks, early Ethernet
Ring	Predictable, no collisions	Single break breaks the network	Token Ring (legacy), SONET
Star	Easy to manage, centralised	Hub/switch is a single point of failure	Most modern LANs (Ethernet)
Mesh	Highly redundant, robust	Expensive, complex cabling	Critical systems, military, backbone

Comparison of Network Topologies

> **Malayalam support:** നെറ്റ്വർക്ക് ടോപ്പോളജികൾ: Bus (single cable), Ring (circle), Star (central hub), Mesh (fully connected). ഓരോന്നിന്റെയും ഗുണങ്ങളും ദോഷങ്ങളും. Star ആണ് ഏറ്റവും സാധാരണമായത് (Ethernet LAN-കളിൽ).

Network Components — Hub, Switch, Router, Gateway

- **Hub:** A simple network device that broadcasts data to all connected devices. Works at Layer 1 (Physical layer) of the OSI model. Inefficient, but inexpensive. Mostly obsolete in modern networks.
- **Switch:** A smarter network device that directs data only to the intended recipient. Works at Layer 2 (Data Link layer). Uses MAC addresses to forward frames. Efficient, widely used.
- **Router:** Connects different networks (e.g., LAN to WAN). Works at Layer 3 (Network layer). Uses IP addresses to route packets. Determines the best path for data.
- **Gateway:** A device that connects networks with different protocols. Translates between different architectures. Works at multiple OSI layers. Example: A router that connects a LAN to the internet is also a gateway.
- **Other Components:** Repeater (amplifies signals), Bridge (connects two LANs), Modem (modulates/demodulates).

Device	Layer	Operation	Intelligence	Example Use
Hub	Physical	Broadcasts to all	None (dumb)	Legacy networks
Switch	Data Link	Forwards to specific MAC	Smart	Modern LANs
Router	Network	Routes between networks	Very smart	Internet connectivity

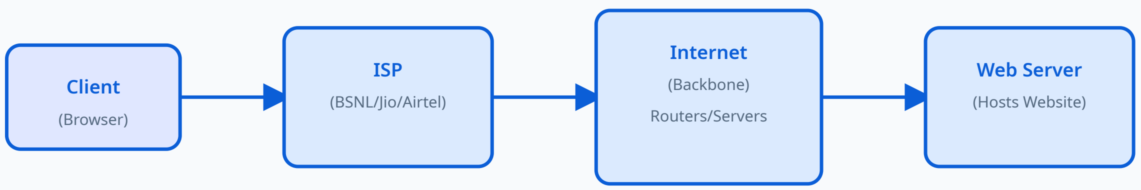
Comparison: Hub, Switch, Router

> **Malayalam support:** നെറ്റ്വർക്ക് ഘടകങ്ങൾ: Hub (broadcast to all), Switch (directs to specific device using MAC), Router (connects different networks using IP), Gateway (protocol translation). Hub-നേക്കാൾ Switch-ഉം Router-ഉം കൂടുതൽ intelligent ആണ്.

Internet Basics — ISP, Web Server, Browser, IP Address, Domain Names, Intranet

- **ISP (Internet Service Provider):** A company that provides internet access to customers. Examples: BSNL, Jio, Airtel, Verizon.
- **Web Server:** A computer that hosts websites and serves web pages to clients. Responds to HTTP/HTTPS requests.
- **Web Browser:** A software application used to access and view web pages. Examples: Chrome, Firefox, Edge, Safari.
- **IP Address (Internet Protocol Address):** A unique numerical label assigned to each device on a network. IPv4: 192.168.1.1, IPv6: 2001:0db8:85a3::8a2e:0370:7334.
- **Domain Name:** A human-readable name for an IP address (e.g., google.com, polytech.edu). Resolved by DNS (Domain Name System).
- **Intranet:** A private network within an organisation that uses internet technologies but is restricted to authorised users. Example: A company's internal portal.

Client → ISP → DNS → Web Server → Client



> **Malayalam support:** ഇന്റർനെറ്റ് അടിസ്ഥാനങ്ങൾ: ISP (service provider), Web Server (website host), Browser (client), IP Address (unique ID), Domain Name (human-readable), Intranet (private network). DNS domain names-നെ IP addresses-ലേക്ക് മാറ്റുന്നു.

World Wide Web, Search Engines, Chatbots & Online Stores

- **World Wide Web (WWW):** A system of interlinked documents and resources accessed via the internet. Uses HTTP/HTTPS protocol. Invented by Tim Berners-Lee in 1989.
- **Search Engines:** Tools that search the web and return relevant results. Examples: Google, Bing, Yahoo, DuckDuckGo.
- **Search Engine Tips:**
  - **Keywords:** Use specific words. Example: "diploma computer applications syllabus".
  - **Quotation Marks (" "):** Search for exact phrases. Example: "Kerala Polytechnic revision 2026".
  - **Site Search (site:):** Limit results to a specific website. Example: site:sitttr.kerala.gov.in "computer applications".
  - **Minus (-):** Exclude words. Example: "computer applications -management".
- **Chatbots:** AI-powered programs that simulate human conversation. Examples: ChatGPT, Gemini, Copilot. Used for customer service, education, and assistance.
- **Online Stores (E-commerce):** Websites where products and services are bought and sold. Examples: Amazon, Flipkart, eBay, GoDaddy (domain/hosting).
- **Amazon Cloud Storage:** Amazon Web Services (AWS) provides cloud storage solutions (S3, EC2). Enables file storage, backup, and web hosting in the cloud.

> ⚠️ **Note:** The syllabus mentions "Basics of Amazon Cloud Storage" and "GoDaddy". These are introduced as examples of cloud services and domain/hosting providers. Amazon S3 (Simple Storage Service) and GoDaddy (domain registration and web hosting) are industry examples used for educational illustration.

> **Malayalam support:** WWW — HTTP/HTTPS വഴി ആക്സസ് ചെയ്യുന്ന പരസ്പരം ബന്ധിപ്പിച്ച ഡോക്യുമെന്റുകളുടെ സംവിധാനം. Search Engines (Google, Bing) keywords, quotation marks, site: search എന്നിവ ഉപയോഗിച്ച് ഫലപ്രദമായി ഉപയോഗിക്കാം. Chatbots (AI), Online Stores (e-commerce), Amazon Cloud Storage (S3, EC2) എന്നിവ ആധുനിക ഇന്റർനെറ്റിന്റെ ഭാഗമാണ്.

🔧 Practical: Network Troubleshooting, Components & Commands

This practical session focuses on basic network troubleshooting, demonstrating network components, printer connection, and using network commands.

Basic Network Troubleshooting

1. **Check Physical Connections:** Ensure cables are properly connected, lights on the switch/router are on.
2. **Restart Devices:** Power cycle the modem, router, and computer.
3. **Check IP Address:** Use `ipconfig` (Windows) or `ifconfig` / `ip a` (Linux) to verify network configuration.
4. **Ping Command:** Use `ping` to test connectivity to another device.
   - `ping 192.168.1.1` — test connectivity to the router.
   - `ping google.com` — test internet connectivity.
5. **Check Network Adapter:** Ensure the network adapter is enabled.
6. **Check DNS Settings:** Use `nslookup` to test DNS resolution.

Network Commands

- **ipconfig (Windows):** Displays IP address, subnet mask, default gateway.
- **ifconfig / ip a (Linux):** Similar to ipconfig, shows network interface information.
- **ping:** Tests connectivity to a host.
- **tracert (Windows) / traceroute (Linux):** Traces the path to a destination.
- **nslookup:** Queries DNS for domain name resolution.
- **netstat:** Shows network connections and routing tables.

Demonstrating Network Components

- **Client Computers:** Workstations that request resources from the network.
- **Switch / Hub:** Interconnect multiple devices in a LAN.
- **Router:** Connects the LAN to the internet (or other networks).
- **Network Cables:** UTP, fibre optic, coaxial — connect devices physically.
- **Printer Connection:** Connect a network printer via IP address or shared printer. Demonstrate how to add a network printer and print a test page.
- **Internet Sharing:** Share an internet connection across multiple devices using a router or software-based ICS.

📘 **Example: Finding Your IP Address**

Windows: Open Command Prompt → type `ipconfig` → look for "IPv4 Address".
Linux: Open Terminal → type `ip a` → look for "inet" under the active interface (e.g., eth0, wlan0).
Example output (Windows):

```
IPv4 Address. . . . . . . . . . . : 192.168.1.100
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . : 192.168.1.1
```

🛡️ **Safety note:** Do not run network troubleshooting commands on networks you do not own or have permission to access. Use `ping` responsibly — it should only be used for diagnostic purposes.

Malayalam support: നെറ്റ്വർക്ക് ട്രബിൾഷൂട്ടിംഗ്: physical connections, restart devices, IP address check (ipconfig/ifconfig), ping, tracert, nslookup. Network components: client computers, switch/hub, router, network cables, printer connection. പ്രായോഗികമായി network connectivity പരിശോധിക്കുകയും search engines ഫലപ്രദമായി ഉപയോഗിക്കുകയും ചെയ്യുക.

💡 **Module IV Summary:** You have learned about computer networks, advantages, types (PAN, LAN, MAN, WAN), topologies (bus, ring, mesh, star), components (hub, switch, router, gateway), internet basics (ISP, web server, browser, IP, domain, intranet), WWW, search engines, chatbots, online stores, cloud storage, and practical troubleshooting techniques.

QUICK REFERENCE

Formula Bank

Key formulas, units, and equations from all modules — useful for revision and problem-solving.

Speed = Distance / Time

Data transfer rate (bps)

Total Sales = Σ (Qty × Price)

Aggregate sales calculation

Average = Total / Count

Mean of data points

Bandwidth = Data Size / Time

Network capacity (bps)

IP Address: 192.168.x.x

Private IPv4 address range

Ping = Request / Response

Network round-trip time (ms)

QUICK VISUAL REFERENCE

Diagram Library

Key diagrams from each module for rapid revision.

📘 Module I

Computer block diagram, memory hierarchy, flowchart symbols, flowchart for adding two numbers.

📗 Module II

OS layers diagram, installation flow, Windows/Linux boot screens (conceptual).

📙 Module III

WiFi network setup, analog vs digital signals, guided media (twisted pair, coaxial, fibre).

📕 Module IV

Star topology, ring topology, internet flow diagram, hub/switch/router comparison.

PRACTICE & EXPLORE

Self-Learning Activities

Official self-learning activities from the syllabus — complete these to reinforce your understanding.

Module I Activities

Week 1 — Class summary + Mind map

Draw a mind map showing characteristics of computers with real-life examples (speed → ATM, accuracy → billing). Write record books.

⏱️ 5.5 hours

Week 2 — Compare memory types

Prepare a table comparing RAM, ROM, Cache, Secondary Memory (speed, volatility, use).

⏱️ 5.5 hours

Week 3 — Classify hardware & software

List all hardware devices at home and classify as input/output/storage. Classify software you use daily into system, application, utility.

⏱️ 5.5 hours

Week 4 — Compare OS & create flowcharts

Compare Windows vs Linux or MS Office vs LibreOffice. Write algorithms and draw flow charts for sample problems.

⏱️ 5.5 hours

Module II Activities

Week 5 — OS functions & examples

Match OS functions with real examples (process management → multitasking). Prepare module notes.

⏱️ 5.5 hours

Week 6 — OS types chart

Prepare a chart showing batch, time-sharing, real-time OS. Write record books.

⏱️ 5.5 hours

Week 7 — Compare mobile OS

Compare Android and iOS (features, openness, apps).

⏱ 5.5 hours

Week 8 — Module summary

Summarise Module II, write record books.

⏱ 5.5 hours

Module III Activities

Week 9 — Input → Process → Output → Storage

Draw the IPOS cycle with examples. Class summary presentation. Write record books.

⏱ 5.5 hours

Week 10 — File organisation

Organise files on your computer into folders and note the benefits.

⏱ 5.5 hours

Week 11 — Database identification

Identify where databases are used (bank, hospital, railway). Differentiate file system vs DBMS.

⏱ 5.5 hours

Week 12 — Communication modes & media

Prepare a table comparing simplex, half-duplex, full-duplex with examples. Compare twisted pair, coaxial, fibre, radio, microwave. Illustrate unguided media.

⏱ 5.5 hours

Module IV Activities

Week 13 — Network types chart

Prepare a chart showing LAN, MAN, WAN with examples. Write record books.

⏱ 5.5 hours

Week 14 — Identify network components

Identify router, switch, modem at home or institution. Prepare module notes.

⏱️ 5.5 hours

Week 15 — Internet services & module summary

List 5 educational uses of the internet and 5 services (email, WWW, cloud). Prepare module summary. Write record books.

⏱️ 5.5 hours

📝 **Total self-learning hours:** 82.5 hours across 15 weeks, as per the official syllabus.

EXAM STRUCTURE

Assessment Methodology

Official CIE and ESE assessment scheme, series test pattern, and evaluation criteria.

Continuous Internal Evaluation (CIE) — 40 Marks

Mode	Description	Marks
Attendance	Class attendance and punctuality	5
CA1	Average of self-learning activities from each module	10
CA2	Continuous evaluation	5
CA3	Practical test (first half of experiments)	10
CA4	Practical test (second half of experiments)	10
CA5	Series Exam (2 modules)	10
Total CIE		**40**

CIE assessment components

End Semester Examination (ESE) — 60 Marks

Mode	Description	Marks
Theory	Written exam (theory portion)	20
Practical	Lab examination (internal)	40
Total ESE		**60**

ESE components

Note: CIE + ESE = 100 marks total. The practical component carries significant weight (40 marks), reflecting the practicum nature of the course.

Series Examination Pattern (Theory)

Part	Type	Marks
Part A	Very short answer (1 mark each)	4 × 1 = 4
Part B	Short answer (3 marks each)	4 × 3 = 12
Part C	Long answer (7 marks each) — with choice	2 × 7 = 14
Total		**25**

Series exam (2 hours) — 25 marks

ESE Question Paper Pattern (Theory — 2.5 hours)

Part	Type	Questions	Marks
Part A	1 mark each	2 from each module (total 8)	8 × 1 = 8
Part B	3 marks each (2 out of 3 per module)	2 × 4 = 8	8 × 3 = 24
Part C	7 marks each (1 set with choice per module)	1 × 4 = 4	4 × 7 = 28
Total		20 questions	**60**

ESE theory paper — 20 marks total

Note: The ESE theory paper is part of the 60-mark ESE total (the other 40 marks come from the practical/lab examination).

Practical Test Evaluation Criteria (40 marks)

Continuous Evaluation (CIE Practical)

- Preparation & Punctuality — 10
- Experimental Setup & Procedure — 10
- Observation & Data Recording — 5
- Analysis & Result Interpretation — 10
- Viva Voce / Concept Understanding — 10
- Workmanship, Discipline & Teamwork — 5
- **Total: 50** (converted to 10 marks as per scheme)

Practical Test (ESE Practical)

- Write-up / Procedure — 10
- Experiment Setup & Execution — 10
- Observation & Result / Output — 8
- Viva Voce — 8
- Record Completion & Neatness — 4
- **Total: 40** (directly added to ESE)

TEST YOURSELF

Module-wise Questions with Answers

Practice questions for each module, with answers provided. Use these for self-assessment and exam preparation.

Module I — Basics of Computer System

Q: What is a computer? List its main characteristics.

A: A computer is an electronic device that processes data according to a set of instructions to produce meaningful information. **Characteristics:** Speed, Accuracy, Storage, Versatility, Diligence, and Automation.

Q: Explain the generations of computers with examples.

A: First Generation (Vacuum Tubes): ENIAC, UNIVAC — large, expensive, high power consumption.
Second Generation (Transistors): IBM 1401 — smaller, more reliable.
Third Generation (ICs): IBM 360 — multiple transistors on one chip.
Fourth Generation (Microprocessors): IBM PC, Apple Mac — personal computers.
Fifth Generation (AI): AI systems, quantum computing — parallel processing and intelligence.

Q: What is the difference between RAM and ROM?

A: RAM (Random Access Memory): Volatile — loses data when power is off. Used for running programs. Can be written to.
ROM (Read Only Memory): Non-volatile — retains data without power. Contains firmware (BIOS). Typically read-only (though some types can be written with special procedures).

Q: Define Algorithm and Flowchart. Draw a flowchart to find the sum of two numbers.

A: Algorithm: A step-by-step procedure to solve a problem.
Flowchart: A graphical representation of an algorithm.
Flowchart for sum of two numbers: Start → Read A, B → SUM = A + B → Print SUM → Stop.
(Refer to the diagram in Module I for visual representation.)

Module II — Operating Systems

Q: What is an Operating System? List its main functions.

A: Operating System: System software that acts as an intermediary between hardware and the user. **Functions:** Process Management, Memory Management, File System Management, Device Management, Security & Protection, User Interface, Resource Allocation.

Q: **Explain the types of operating systems based on processing technique.**

A: **Batch Processing OS:** Jobs are processed in batches with no user interaction.
Time-Sharing OS: Multiple users share CPU time in small time slices (interactive).
Real-Time OS: Responds to events in a guaranteed time frame (used in embedded systems).

Q: **What is the difference between multiprogramming, multitasking, and multiprocessing?**

A: **Multiprogramming:** Multiple programs are loaded into memory, and the CPU switches between them to stay busy.
Multitasking: Similar but includes user interaction — multiple tasks appear to run simultaneously.
Multiprocessing: Uses multiple CPUs to execute multiple processes simultaneously (requires special hardware).

Q: **Compare Android and iOS as mobile operating systems.**

A: **Android:** Open-source (based on Linux), highly customisable, large app ecosystem, supports multiple vendors.
iOS: Closed-source (Apple), highly secure, curated app store, seamless ecosystem, regular updates.

Module III — Data Processing & Communication

Q: **What is data processing? List its types.**

A: **Data Processing:** The process of collecting, manipulating, and converting raw data into meaningful information. **Types:** Batch Processing, Real-Time Processing, Online Processing, Distributed Processing.

Q: **Explain the elements of a data communication system.**

A: **Sender (Source):** Device that sends the data.
Receiver (Destination): Device that receives the data.
Message: The data being transmitted.
Transmission Medium: The physical path (wired or wireless).
Protocol: A set of rules governing the communication.

Q: **What are the modes of data communication? Give examples.**

A: **Simplex:** One-way communication — Radio broadcast, TV transmission.
Half-Duplex: Two-way but only one at a time — Walkie-talkie.
Full-Duplex: Two-way simultaneously — Telephone, video conferencing.

Q: **Compare guided and unguided transmission media with examples.**

A: Guided (Wired): Twisted Pair (UTP/STP), Coaxial Cable, Optical Fibre. Provide a physical path, higher bandwidth, more secure.
Unguided (Wireless): Radio waves, Microwaves, Infrared. No physical path, less secure, more flexible.

Module IV — Computer Networks & Topologies

Q: **What is a computer network? List its advantages.**

A: Computer Network: A set of interconnected computers and devices that communicate with each other. **Advantages:** Resource Sharing, Communication, Centralised Data Management, Cost Efficiency, Reliability, Scalability, Remote Access, Security.

Q: **Explain the types of networks: PAN, LAN, MAN, WAN.**

A: PAN: Personal Area Network — few meters (Bluetooth).
LAN: Local Area Network — building/campus (Ethernet, WiFi).
MAN: Metropolitan Area Network — city (cable TV network).
WAN: Wide Area Network — country/global (the Internet).

Q: **Compare star, ring, bus, and mesh topologies.**

A: Star: All devices connected to a central hub — easy to manage, single point of failure.
Ring: Each device connected to two others in a ring — predictable, one break breaks the network.
Bus: Single central cable — simple, low cost, single point of failure.
Mesh: Every device connected to every other — highly redundant, robust, expensive.

Q: **What is the difference between a hub, a switch, and a router?**

A: Hub: Broadcasts data to all devices (Layer 1 — Physical). Inefficient.
Switch: Forwards data to the specific device using MAC address (Layer 2 — Data Link). Efficient.
Router: Connects different networks using IP address (Layer 3 — Network). Routes data between networks.

PRACTICE EXAM

Model Question Paper

A practice paper based on the official pattern. **Note:** This is a practice model prepared for study purposes, not an official SITTTR question paper.

Course: Computer Applications in Management (1254) | **Semester:** I | **Time:** 2.5 hours | **Max Marks:** 60

Part A: 8 × 1 = 8 marks | Part B: 8 × 3 = 24 marks | Part C: 4 × 7 = 28 marks

Part A — Very Short Answer (1 mark each)

Answer any 8 questions. Each question carries 1 mark.

1. What is a computer?
2. Define RAM and ROM.
3. What is an Operating System?
4. List any two functions of an OS.
5. What is data processing?
6. What is a simplex communication mode?
7. What is a LAN?
8. Name any two network topologies.
9. What is a router?
10. What is an IP address?

Part B — Short Answer (3 marks each)

Answer any 8 questions. Each question carries 3 marks.

1. Explain the characteristics of computers.
2. Explain the generations of computers.
3. Explain the layers of an operating system.
4. What is the difference between multiprogramming and multitasking?
5. Explain the elements of a data communication system.
6. What are the modes of data communication? Give examples.
7. Explain the types of networks: PAN, LAN, MAN, WAN.
8. Compare star and ring topologies.
9. What is the difference between a hub and a switch?
10. Explain the basics of the Internet (ISP, web server, browser).

Part C — Long Answer (7 marks each)

Answer any 4 questions. Each question carries 7 marks. (One set with choice per module.)

1. **Module I:** Explain the components of a computer system with a block diagram. OR Explain the types of computer memory with a hierarchy diagram.
2. **Module II:** Explain the functions of an operating system in detail. OR Compare Windows and Linux operating systems.
3. **Module III:** Explain the types of data transmission media (guided and unguided) with examples and diagrams. OR Explain the elements of a data communication system and the modes of data communication.
4. **Module IV:** Explain network topologies (bus, ring, star, mesh) with diagrams, advantages, and disadvantages. OR Explain network components (hub, switch, router, gateway) and their functions.

💡 **Practice tip:** Attempt this paper under timed conditions. Use the Q&A section for reference. Check your answers against the module content.

LAST-MINUTE REVIEW

Quick Revision

Condensed key points from each module, common mistakes, and an exam checklist.

📘 Module I — Computer Basics

- **Characteristics:** Speed, Accuracy, Storage, Versatility, Diligence, Automation.
- **Generations:** Vacuum tubes → Transistors → ICs → Microprocessors → AI.
- **Memory:** RAM (volatile), ROM (non-volatile), Cache, HDD, SSD.
- **Software:** System (OS, drivers) vs Application (MS Office, browsers).
- **Languages:** Machine, Assembly, High-Level (C, Python).
- **Translators:** Compiler (whole program) vs Interpreter (line by line).

📗 Module II — Operating Systems

- **Functions:** Process, Memory, File, Device, Security, UI, Resource.
- **Types:** Batch, Time-Sharing, Real-Time.
- **Techniques:** Multiprogramming, Multitasking, Multiprocessing.
- **Popular OS:** Windows (GUI, proprietary), Linux (open-source, stable).
- **Mobile OS:** Android (open), iOS (closed), features: touch, apps, cloud.

📙 Module III — Data Processing & Communication

- **Data Processing:** Batch, Real-Time, Online, Distributed.
- **File Types:** Master, Transaction, Reference, Report.
- **DBMS:** Reduces redundancy, ensures consistency, uses SQL.
- **Communication Modes:** Simplex (one-way), Half-Duplex (alternate), Full-Duplex (simultaneous).
- **Transmission Media:** Guided (Twisted Pair, Coaxial, Fibre) vs Unguided (Radio, Microwave, Infrared).
- **Modem:** Modulates (digital → analog) and Demodulates (analog → digital).

📕 Module IV — Computer Networks

- **Types:** PAN (few meters), LAN (building), MAN (city), WAN (global).
- **Topologies:** Bus (single cable), Ring (circle), Star (central hub), Mesh (fully connected).
- **Components:** Hub (broadcast), Switch (MAC address), Router (IP address), Gateway (protocol conversion).
- **Internet:** ISP, Web Server, Browser, IP Address, Domain Name, Intranet.

- **Commands:** ipconfig/ifconfig, ping, tracert/traceroute, nslookup.
- **Cloud:** Amazon S3, EC2 — storage and computing in the cloud.

⚠️ Common Mistakes to Avoid

Confusing RAM and ROM: RAM is volatile, ROM is non-volatile. Remember: RAM can be written to; ROM is typically read-only.

Mixing up topologies: Star has a central hub; Ring forms a circle; Bus has a single cable; Mesh has multiple connections.

Hub vs Switch vs Router: Hub broadcasts to all, Switch uses MAC addresses, Router uses IP addresses.

Compiler vs Interpreter: Compiler translates the whole program; Interpreter translates line by line.

✅ Exam Checklist

- ☐ I know the characteristics and generations of computers.
- ☐ I can explain the components of a computer system and the memory hierarchy.
- ☐ I understand the functions and types of operating systems.
- ☐ I can differentiate between multiprogramming, multitasking, and multiprocessing.
- ☐ I know the types of data processing and file management.
- ☐ I understand the elements of data communication and transmission media.
- ☐ I can explain network topologies and compare them.
- ☐ I know the functions of hub, switch, router, and gateway.
- ☐ I understand Internet basics (ISP, web server, browser, IP, domain).
- ☐ I can use ipconfig/ifconfig, ping, and traceroute commands.
- ☐ I have practiced the model paper and reviewed the Q&A.

KEY TERMS

Glossary

Important technical terms and their one-line meanings.

Term	Meaning
CPU	Central Processing Unit — the "brain" of the computer
RAM	Random Access Memory — volatile memory for running programs
ROM	Read Only Memory — non-volatile memory for firmware

Term	Meaning
HDD	Hard Disk Drive — magnetic storage for permanent data
SSD	Solid State Drive — flash-based storage, faster than HDD
OS	Operating System — manages hardware and provides a platform for applications
GUI	Graphical User Interface — visual way to interact with a computer
CLI	Command Line Interface — text-based way to interact with a computer
DBMS	Database Management System — software for managing structured data
SQL	Structured Query Language — language for querying databases
LAN	Local Area Network — network within a limited area
WAN	Wide Area Network — network covering a large geographic area
PAN	Personal Area Network — network for personal devices
MAN	Metropolitan Area Network — network covering a city
IP Address	Internet Protocol Address — unique identifier for a device on a network
DNS	Domain Name System — translates domain names to IP addresses
ISP	Internet Service Provider — company that provides internet access
HTTP	HyperText Transfer Protocol — protocol for web communication
HTTPS	HTTP Secure — secure version of HTTP
FOSS	Free and Open Source Software — software with freely available source code
BIOS	Basic Input Output System — firmware that starts the computer
SMPS	Switched Mode Power Supply — power supply for computers
CMOS	Complementary Metal-Oxide-Semiconductor — battery-powered memory for system settings

Glossary of important terms

FURTHER READING

References & Resources

Textbooks, reference books, and web resources listed in the official syllabus.

📚 Textbooks

1. Rajaraman — *Fundamentals of Computers*, Prentice Hall of India, 2003.
2. P.K Sinha, Priti Sinha — *Computer Fundamentals*, Kalyani Publishers, 2nd Edition.
3. Shelly — *Business Data Communication*, Thomson Learning, Bombay, 2003.
4. White — *Data Communication & Computer Network*, Thomson Learning, Bombay, 2012.

📖 Reference Books

1. B. Ram — *Computer Fundamentals*, New Age International, 2000.
2. Behrouz A. Forouzan — *Data Communications and Networking*, McGraw Hill Education, 2017.

🌐 Web Resources (from syllabus)

- Fortinet — Wireless Network (Module III)
- Lenovo — What is Modem (Module III)
- Network Cable Crimping and Testing Tools (Module IV)